ENGS106-partc-writeup

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March 2026

1 Brief Description

The key idea behind the diffusion model is learning to predict noise and removing noise at different steps to unveil the original image.

We encode with sin/cos frequency embeddings and use a U-Net architecture for noise prediction. When removing noise, we assign greater weights at earlier time steps($t = 999$) when the image is noisier and lower weights as t approaches 0.

2 Evaluation

Image placement is weird. Refer to bottom for images.

3 Reflections

This was a chill coding assignment. Much of it was just understanding PyTorch documentation/syntax and copying the formulas provided.

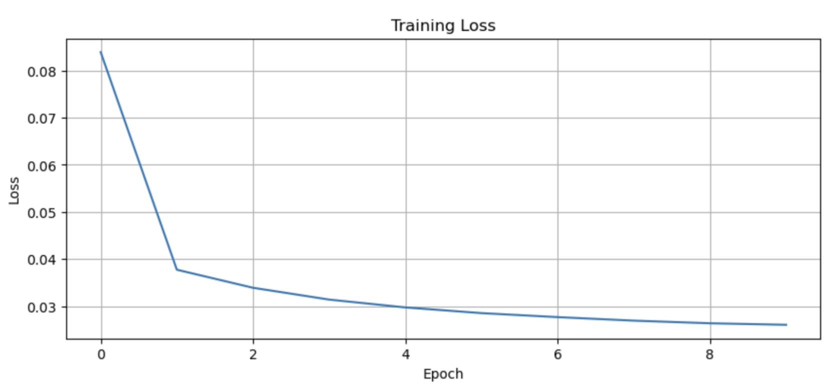

Figure 1: Enter Caption

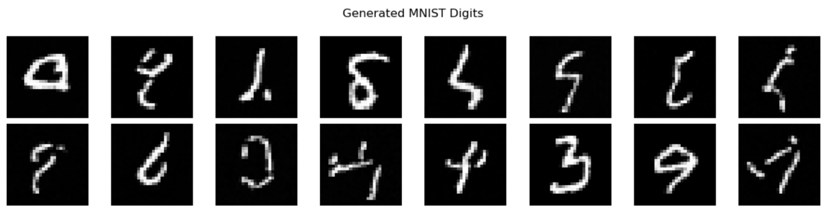

Figure 2: Enter Caption

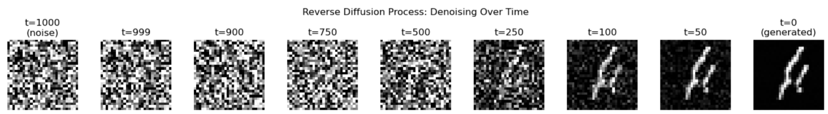

Figure 3: Enter Caption

I am surprised by how good models can detect noise though. If you were to provide an image at $t = 999$, it appears to be pure noise to the human eye but to the model, it's able to relatively well recover the original image.